

Deep learning semantic segmentation for land use and land cover classification at very high spatial resolution — a review of data sources and methods

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Abstract

Land use/land cover (LULC) classification is a critical task in remote sensing analysis with applications in many domains, including agriculture, urban planning, and environmental monitoring. Recent innovations in remote sensing have enabled wider access to very high spatial resolution (VHSR) data, but classical models used for LULC classification at moderate spatial resolutions tend to perform poorly on this VHSR data. Deep learning-driven semantic segmentation approaches to LULC classification are becoming increasingly popular due to their capacity to leverage spatial information in addition to spectral signatures without needing hand-crafted features, thus addressing issues that arise from increased intra-class variance at VHSR and reducing the amount of human oversight needed to build classifiers. This review paper aims to provide a comprehensive overview of recent semantic segmentation approaches for LULC classification at sub 5-meter spatial resolution with a focus on multispectral imagery. We catalog common sensor platforms and data sources for obtaining data at VHSR as well as popular datasets for training and evaluating LULC models using VHSR data, along with a survey of methods for LULC classification at VHSR that utilize deep learning-driven semantic segmentation. Finally, key areas of emerging research that have implications on the future of LULC classification are identified, including Transformer models and self-supervised pre-training schemas.

Keywords: land use and land cover classification, very high spatial resolution, semantic segmentation, deep learning, remote sensing

1 Introduction

Land use and land cover (LULC) data provides information about the biophysical properties of Earth’s surface and the human activities centered around said properties, making them an essential component of environmental monitoring, natural resource management, and urban planning with far-reaching applications in agriculture, hydrology, and forestry (Chen et al., 2019a; Crippa et al., 2021; Friedlingstein et al., 2019; Grekousis et al., 2015; Loveland et al., 2000; Townshend, 1992; Wulder et al., 2018). In the case of land cover, these maps demarcate regions on Earth’s surface according to a consistent ontology of classes that correspond to natural resources and human developments (i.e., forest, grassland, developed, cropland, structures, etc.), while land use maps provide spatial information on how humans interact with said resources and developments (i.e., residential areas, timber stands, parking lots, public areas, etc.) (Anderson, 1971, 1976; Foley et al., 2005; Huang et al., 2018a; Vali et al., 2020). These products are pivotal to understanding and extracting information about Earth’s resources and monitoring human impact (Brovkin et al., 2004; Crippa et al., 2021; Dale et al., 1998; de Groot et al., 2010; Friedlingstein et al., 2019; Lambin et al., 2001; Meyer and Turner, 1996; Pielke Sr. et al., 2011; Rockström et al., 2009; Turner et al., 1990; Weiss et al., 2020). Consequently, the creation of accurate LULC maps for analysis has historically been a prominent area of investigation in remote sensing research, with a large volume of scientific research focused on leveraging various data sources, methods, and applications with the goal of producing accurate

and insightful LULC maps as efficiently as possible (Anderson, 1977; Cadenasso et al., 2007; Cihlar, 2000; Friedl et al., 2002; Maxwell et al., 2018; Phiri and Morgenroth, 2017).

Over the past two decades, the availability of optical reflectance data at very high spatial resolution (VHSR) has grown substantially due to the launch of numerous commercial Earth observation (EO) satellites (Belward and Skøien, 2015). For this paper, VHSR is defined as any data captured from an aerial or orbital platform with a spatial resolution at or below 5 meters (5 m). This VHSR data has enabled the creation of LULC maps with a high level of detail, as shown in Figure 1. However, there are several challenges associated with the use of VHSR data for analysis: the increased spatial resolution of the data leads to increased noise, variation in spectral responses for similar objects, and more diverse scenes as small scale objects that are indiscernible at coarse resolutions become more apparent at higher resolutions (Blaschke, 2010; Strahler et al., 1986; Tong et al., 2020b; Woodcock and Strahler, 1987). This increased complexity in VHSR data presents unique analytical challenges that must be addressed in order to create LULC classifiers that are accurate, reliable, and generalizable (Balha et al., 2021; Diakogiannis et al., 2020; Hussain et al., 2013; Tehrany et al., 2014).

One trend that has become prevalent in the past decade is the use of deep learning-based models for LULC classification as opposed to classical machine learning models. These models have demonstrated significant value in addressing challenges associated with remote sensing data analysis, particularly in cases where traditional machine learning approaches are ineffective, as they are well-suited for modeling complex,